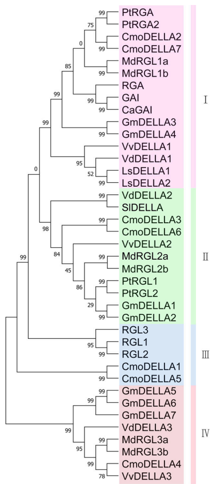
REACTR Validation Summary Table
Nathan Lee

	Documented phylogenetic tree gene family size (all species in paper)	Gene family size (only target and base)	% of documented orthologs also found by REACTR	False/Truncated isoforms detected	Robinson-Foulds distance of documented phylogenetic trees vs cleaned REACTR trees
DELLA: Vaccinium darrowii	39	8	100%	0	0
PHY: Oryza Sativa	20	8	100%	1	0
HOOKLESS: Solanum lycopersicum	8	4	100%	3	0

Methodology
REACTR was run on the base and target species, with the documented base species's gene family FASTAs as the query. The Newick string and image of the REACTR generated tree were downloaded. The locus tags of the protein sequences were then matched and renamed to fit the standardized protein symbols (i.e. RGL3) using UniProt (The Uniprot Consortium, 2025) and NCBI BLAST (Altschul et al., 1990) searches. The Newick tree was then cleaned to remove false/truncated isoforms.[1] Images of phylogenetic trees were downloaded from previous literature[2] The downloaded images were uploaded to Treemble (Allard & Kumar, 2026) to generate a Newick string. The Newick string was exclusively composed of the base and target species' gene families. The Newick string was uploaded to iTol (Letunic & Bork, 2024) to confirm that the Newick string was accurate to the phylogenetic tree from previous literature. The two Newick strings, from REACTR and previous literature (transformed through Treemble), were submitted to T-REX (Boc et al., 2012). The Robinson-Foulds calculated distance and its corresponding matrix was documented.

[1] False/truncated isoforms were removed for manual cleaning during validation. REACTR can pick up false isoforms during its homology-domain search, causing such isoforms to appear in downstream analysis like phylogenetic tree generation. Such isoforms are the effect of errors during genome assembly and reflect REACTR's sensitivity. However, isoforms can be removed manually through examination of phylogenetic trees. REACTR does not automatically remove isoforms due to a prioritization of sensitivity and emphasis on discovering complex, evolutionary events. Such designs are not uncommon amongst established tools, like Orthofinder (Emms et al., 2026).
[2] Retrieved from Zhou et al., 2024 (DELLA: Vaccinium darrowii), Schrager-Lavelle et al., 2016 (PHY: Oryza Sativa), Chaabouni et al., 2016 (HOOKLESS: Solanum lycopersicum)

DELLA: Vaccinium darrowii

<table>
<tr><th>Previous Literature Phylogenetic Tree (Zhou et al., 2024)</th><th>Treemble Cleaned Phylogenetic Tree</th><th>ITol Cleaned from Treemble String</th></tr>
<tr>
<td>
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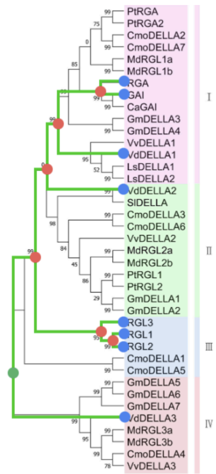
<td>
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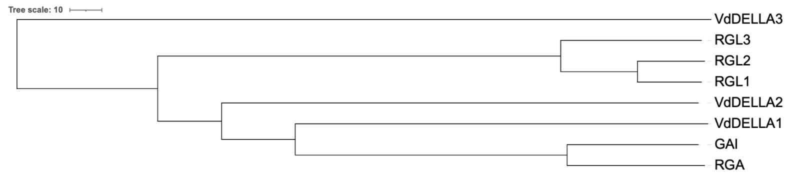
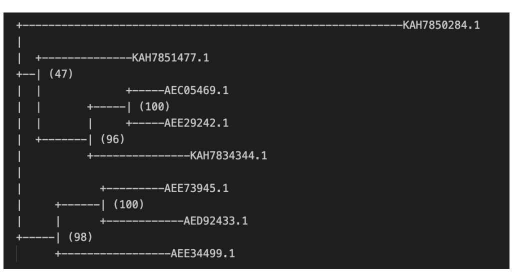
<td>
</td>
</tr>
</table>

Treemble Generated Newick	REACTR Original Newick	REACTR Cleaned Newick (Tag names, identity matching)	Robinson-Foulds Distance (T-REX)
(((((RGA:43,GAI:41):85,VdDELLA1:129):23,VdDELLA2:149):20,(RGL3:44,(RGL1:20,RGL2:21):24):126):44,VdDELLA3:217):0;	(KAH7850284.1:1.1597499983,(KAH7851477.1:0.2943587276,((AEC05469.1:0.1246721418,AEE29242.1:0.1306299231)100:0.1324360415,KAH7834344.1:0.3240698106)96:0.1688512649)47:0.0250594701,((AEE73945.1:0.1981568522,AED92433.1:0.2674184850)100:0.1497274165,AEE34499.1:0.3616562861)98:0.1169479110);	(VdDELLA3:1.1597499983,(VdDELLA2:0.2943587276,((RGA:0.1246721418,GAI:0.1306299231)100:0.1324360415,VdDELLA1:0.3240698106)96:0.1688512649)47:0.0250594701,((RGL1:0.1981568522,RGL2:0.2674184850)100:0.1497274165,RGL3:0.3616562861)98:0.1169479110);	0

<table>
<tr><th>REACTR Tree (IQ-Tree, pre-cleaning)</th><th>T-REX Distance Matrix (Robinson Foulds Validation)</th></tr>
<tr>
<td>
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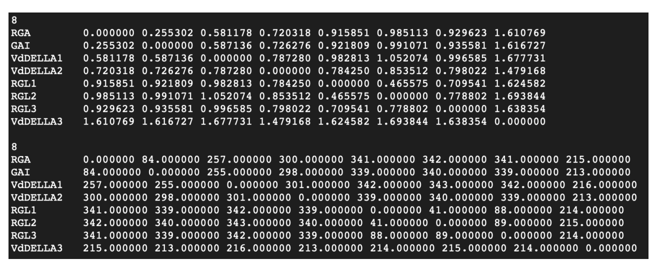
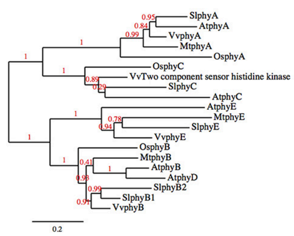
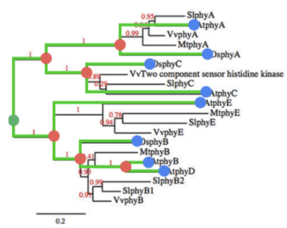
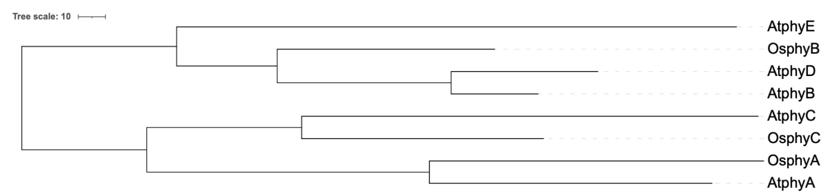
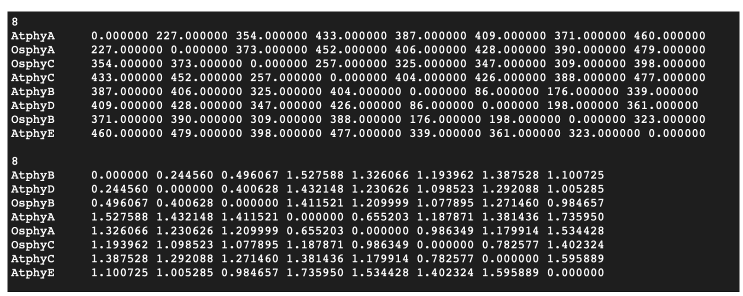
<td>
<table>
<tr><td>8</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>RGA</td><td>0.000000</td><td>0.255302</td><td>0.581178</td><td>0.720318</td><td>0.915851</td><td>0.985113</td><td>0.929623</td><td>1.610769</td></tr>
<tr><td>GAI</td><td>0.255302</td><td>0.000000</td><td>0.587136</td><td>0.726276</td><td>0.921809</td><td>0.991071</td><td>0.935581</td><td>1.616727</td></tr>
<tr><td>VdDELLA1</td><td>0.581178</td><td>0.587136</td><td>0.000000</td><td>0.787280</td><td>0.982813</td><td>1.052074</td><td>0.996585</td><td>1.677731</td></tr>
<tr><td>VdDELLA2</td><td>0.720318</td><td>0.726276</td><td>0.787280</td><td>0.000000</td><td>0.784250</td><td>0.853512</td><td>0.798022</td><td>1.479168</td></tr>
<tr><td>RGL1</td><td>0.915851</td><td>0.921809</td><td>0.982813</td><td>0.784250</td><td>0.000000</td><td>0.465575</td><td>0.709541</td><td>1.624582</td></tr>
<tr><td>RGL2</td><td>0.985113</td><td>0.991071</td><td>1.052074</td><td>0.853512</td><td>0.465575</td><td>0.000000</td><td>0.778802</td><td>1.693844</td></tr>
<tr><td>RGL3</td><td>0.929623</td><td>0.935581</td><td>0.996585</td><td>0.798022</td><td>0.709541</td><td>0.778802</td><td>0.000000</td><td>1.638354</td></tr>
<tr><td>VdDELLA3</td><td>1.610769</td><td>1.616727</td><td>1.677731</td><td>1.479168</td><td>1.624582</td><td>1.693844</td><td>1.638354</td><td>0.000000</td></tr>
<tr><td>8</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>RGA</td><td>0.000000</td><td>84.000000</td><td>257.000000</td><td>300.000000</td><td>341.000000</td><td>342.000000</td><td>341.000000</td><td>215.000000</td></tr>
<tr><td>GAI</td><td>84.000000</td><td>0.000000</td><td>255.000000</td><td>298.000000</td><td>339.000000</td><td>340.000000</td><td>339.000000</td><td>213.000000</td></tr>
<tr><td>VdDELLA1</td><td>257.000000</td><td>255.000000</td><td>0.000000</td><td>301.000000</td><td>342.000000</td><td>343.000000</td><td>342.000000</td><td>216.000000</td></tr>
<tr><td>VdDELLA2</td><td>300.000000</td><td>298.000000</td><td>301.000000</td><td>0.000000</td><td>339.000000</td><td>340.000000</td><td>339.000000</td><td>213.000000</td></tr>
<tr><td>RGL1</td><td>341.000000</td><td>339.000000</td><td>342.000000</td><td>339.000000</td><td>0.000000</td><td>41.000000</td><td>88.000000</td><td>214.000000</td></tr>
<tr><td>RGL2</td><td>342.000000</td><td>340.000000</td><td>343.000000</td><td>340.000000</td><td>41.000000</td><td>0.000000</td><td>89.000000</td><td>215.000000</td></tr>
<tr><td>RGL3</td><td>341.000000</td><td>339.000000</td><td>342.000000</td><td>339.000000</td><td>88.000000</td><td>89.000000</td><td>0.000000</td><td>214.000000</td></tr>
<tr><td>VdDELLA3</td><td>215.000000</td><td>213.000000</td><td>216.000000</td><td>213.000000</td><td>214.000000</td><td>215.000000</td><td>214.000000</td><td>0.000000</td></tr>
</table></td>
</tr>
</table>

PHY: Oryza Sativa

Previous Literature Phylogenetic Tree ([Schrager-Lavelle et al., 2016](#))	Treemble Cleaned Phylogenetic Tree	ITol Cleaned from Treemble String
		

Treemble Generated Newick	REACTR Original Newick	REACTR Cleaned Newick	Robinson-Foulds Distance (T-REX)
(((AtphyA:104,OsphyA:123):104,(OsphyC:89,AtphyC:168):57):46,(AtphyE:206,(OsphyB:80,(AtphyB:32,AtphyD:54):64):37):57):0;	(AEE84000.1:0.6545430041,(NP_001389072.1:0.1922894228,(AEE83721.1:0.1677459286,(ANM63140.1:0.0413747359,AEC06808.1:0.0000026845)100:0.0745601759)100:0.1360673694)100:0.1401141572,((NP_001405525.1:0.4283623775,AEE28462.1:0.2268402708)100:0.3323859187,(NP_001405645.1:0.2945056913,AED94021.1:0.4880709750)99:0.1326169304)100:0.3206583108);	(AtphyE:0.6545430041,(OsphyB:0.19,(AtphyB:0.17,AtphyD:0.0745601759)100:0.1360673694)100:0.1401141572,((AtphyA:0.4283623775,OsphyA:0.2268402708)100:0.3323859187,(OsphyC:0.2945056913,AtphyC:0.4880709750)99:0.1326169304)100:0.3206583108);	0

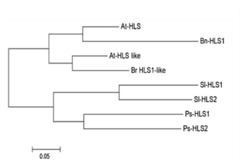
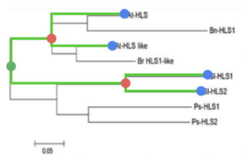
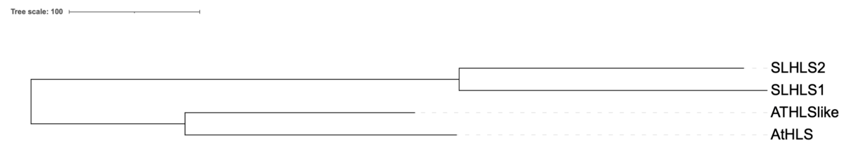
REACTR Tree (IQ-Tree, pre-cleaning)	T-REX Distance Matrix (Robinson Foulds Validation)
	

```
8
AtphyA     0.000000 227.000000 354.000000 433.000000 387.000000 409.000000 371.000000 460.000000
OsphyA     227.000000 0.000000 373.000000 452.000000 406.000000 428.000000 390.000000 479.000000
OsphyC     354.000000 373.000000 0.000000 257.000000 325.000000 347.000000 309.000000 398.000000
AtphyC     433.000000 452.000000 257.000000 0.000000 404.000000 426.000000 388.000000 477.000000
AtphyB     387.000000 406.000000 325.000000 404.000000 0.000000 86.000000 176.000000 339.000000
AtphyD     409.000000 428.000000 347.000000 426.000000 86.000000 0.000000 198.000000 361.000000
OsphyB     371.000000 390.000000 309.000000 388.000000 176.000000 198.000000 0.000000 323.000000
AtphyE     460.000000 479.000000 398.000000 477.000000 339.000000 361.000000 323.000000 0.000000

8
AtphyB     0.000000 0.244560 0.496067 1.527588 1.326066 1.193962 1.387528 1.100725
AtphyD     0.244560 0.000000 0.400628 1.432148 1.230626 1.098523 1.292088 1.005285
OsphyB     0.496067 0.400628 0.000000 1.411521 1.209999 1.077895 1.271460 0.984657
AtphyA     1.527588 1.432148 1.411521 0.000000 0.655203 1.187871 1.381436 1.735950
OsphyA     1.326066 1.230626 1.209999 0.655203 0.000000 0.986349 1.179914 1.534428
OsphyC     1.193962 1.098523 1.077895 1.187871 0.986349 0.000000 0.782577 1.402324
AtphyC     1.387528 1.292088 1.271460 1.381436 1.179914 0.782577 0.000000 1.595889
AtphyE     1.100725 1.005285 0.984657 1.735950 1.534428 1.402324 1.595889 0.000000
```

Removed isoform (ANM63140.1)

HOOKLESS: Solanum lycopersicum

Previous Literature Phylogenetic Tree (Chaabouni et al., 2016)	Treemble Cleaned Phylogenetic Tree	ITol Cleaned from Treemble String
		

Treemble Generated Newick	REACTR Original Newick	REACTR Cleaned Newick	Robinson-Foulds Distance (T-REX)
((AtHLS:208,ATHLSlike:176):118,(SLHLS1:236,SLHLS2:218):328):0;	(XP_004235944.1:1.6291598128,((AED98343.1:0.0000025602,ANM70638.1:0.0000021128)100:0.6049279491,((AEC07401.1:0.0000025602,AEC07402.2:0.0000021128)100:0.0970948845,(AEE86813.1:0.0000026968,ANM66121.1:0.0000021128)99:0.0669425103)99:0.1433370798)95:0.1540259248,(XP_004233456.1:0.2403762388,XP_004231914.1:0.1903798615)52:0.0797280129);	((ATHLSlike:0.0000025602,ATHSL:0.0000026968):0.1540259248,(SLHLS1:0.2403762388,SLHLS2:0.1903798615):0.0797280129);	0

REACTR Tree (IQ-Tree, pre-cleaning)	T-REX Distance

```
+-----------------------------------------------------------XP_004235944.1
|
|                           +--AED98343.1
|    +----------------------| (100)
|    |                      +--ANM70638.1
+----| (95)
|    |        +--AEC07401.1
|    |     +--| (100)
|    |     |  +--AEC07402.2
|    +-----| (99)
|          |  +--AEE86813.1
|          +--| (99)
|             +--ANM66121.1
|
|  +--------XP_004233456.1
+--| (52)
   +-------XP_004231914.1
```

```
4
ATHLSlike    0.000000 0.000005 0.394405 0.344408
ATHSL        0.000005 0.000000 0.394405 0.344408
SLHLS1       0.394405 0.394405 0.000000 0.430756
SLHLS2       0.344408 0.344408 0.430756 0.000000

4
AtHLS        0.000000 384.000000 562.000000 544.000000
ATHLSlike    384.000000 0.000000 530.000000 512.000000
SLHLS1       562.000000 530.000000 0.000000 454.000000
SLHLS2       544.000000 512.000000 454.000000 0.000000
```

Removed three isoforms (ANM70638.1, AEC07402.2, ANM66121.1), removed extra paralog (AEC07401.1) to match query of reference literature